
Attention-based Spatial Feature Fusion Network (ASFF-Net) with Asymmetric Loss Function for Remaining Useful Life Prediction

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Abstract

Accurate prediction of the Remaining Useful Life (RUL) of turbofan engines is critical for aerospace preventive maintenance. Traditional approaches use symmetric loss functions, such as Mean Squared Error (MSE), which penalize early and late predictions equally, ignoring the catastrophic risk associated with overestimating engine life. This paper proposes ASFF-Net (Attention-based Spatial Feature Fusion Network), a deep architecture that integrates: (1) a Dynamic Spatial Attention layer to weight critical sensors; (2) parallel processing pathways combining 1D Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks; and (3) a risk-oriented Asymmetric Loss Function. Experiments on the NASA CMAPSS FD001 dataset demonstrate that ASFF-Net achieves a Mean Absolute Error (MAE) of 13.24 cycles on the test set, outperforming the Random Forest baseline (MAE 13.67) and ensuring conservative predictions that maximize operational safety.

1 Introduction

Data-driven predictive maintenance has transformed industrial fleet management. In the aerospace sector, turbofan engine degradation is a complex and multivariate process. Current literature has explored deep architectures to extract temporal patterns from sensors, but important gaps still remain.

First, standard models treat all sensors with equal importance throughout the entire life cycle, which is unrealistic. Furthermore, purely sequential networks struggle to capture both the long-term degradation trend and abrupt anomaly peaks. Most importantly, symmetric optimization metrics do not reflect the asymmetric cost of the real world, where predicting that an engine will fail later than reality can result in catastrophic failures.

To address these issues, this work proposes ASFF-Net. The three main contributions of this work are:

- **Sensor-Focused Attention:** implementation of a dynamic weighting layer that learns to assign variable importance to specific sensors based on the global temporal context.
- **Parallel Fusion Pathways (LSTM + CNN 1D):** a branched topology where an LSTM models slow degradation, while a CNN detects sudden anomalies in parallel.
- **Asymmetric Loss:** formulation of a customized loss function that severely penalizes late estimates, forcing the model to adopt a conservative behavior focused on physical risk.

2 Related Work

RUL prediction using the CMAPSS dataset has been widely investigated. Pioneering works used classical machine learning techniques, such as Support Vector Regression (SVR) and Random Forests [Breiman, 2001, Li et al., 2018]. With the advent of deep learning, Babu et al. [Babu et al., 2016] proposed the use of one-dimensional Convolutional Neural Networks (CNNs) to extract spatial features from sensors without manual intervention. Zheng et al. [Zheng et al., 2017] and Wu et al. [Wu et al., 2018] expanded this notion by introducing LSTM architectures to model long temporal dependencies. More recently, attention mechanisms have been proposed to focus on relevant temporal features [Chen et al., 2020]. However, the vast majority of these models optimize a symmetric loss (MSE), neglecting the operational risk of overestimating RUL.

3 Methodology

3.1 Data Pipeline and Preprocessing

The pipeline begins with noise removal. From the original 21 sensors of CMAPSS FD001 [Saxena et al., 2008], 9 constant or irrelevant variables are excluded (`config_op_1` to 3, and `sensor_1`, 5, 10, 16, 18, 19), leaving 15 active features. RUL is calculated assuming a piecewise linear degradation, applying an upper limit (clipping) of 130 cycles, focusing the model on the final phase of structural degradation.

The data is normalized using `MinMaxScaler` (scaled between 0 and 1). For temporal processing, the data is restructured through three-dimensional sliding windows, where each sample has the dimension (T, F) , with $T = 30$ being the sequence length and $F = 15$ the number of features.

3.2 Network Architecture (ASFF-Net)

The architecture is instantiated in the `build_asff_net()` function.

1. **Dynamic Spatial Attention:** the customized layer (`DynamicSensorWeights`) receives the three-dimensional input $(None, 30, 15)$ and applies a temporal average (`reduce_mean`) to extract the global context. This vector passes through a dense layer of 16 neurons (ReLU) and another of 15 neurons (*sigmoid*), generating weights that iteratively multiply the original sensors.
2. **Parallel Pathways:** the weighted input is divided into three divergent streams:
 - *LSTM Pathway:* two cascaded LSTM layers (64 and 32 units), with a Dropout layer (rate 0.2) applied after the first LSTM.
 - *CNN Pathway:* a Conv1D layer (32 filters, kernel 3) with ReLU activation, followed by a pooling operation (`GlobalMaxPooling1D`).
 - *Direct Pathway (Skip):* Flatten layer connected to a dense layer (32 units, ReLU).
3. **Fusion and Prediction:** the outputs (32 dimensions each) are joined via `layers.Concatenate()`, resulting in a 96-dimensional vector. This vector passes through a Dropout (rate 0.3), a dense layer (32 units, ReLU) and, finally, a linear neuron in the `rul_output` layer.

3.3 Asymmetric Loss Function and Hyperparameters

The main mathematical differential lies in the `asymmetric_loss()` function. Let $d = \hat{y} - y$ be the residual error. The loss is calculated prioritizing safety:

$$L(d) = \begin{cases} \exp\left(\frac{d}{10}\right) - 1, & d > 0 \text{ (late/dangerous)} \\ \exp\left(-\frac{d}{13}\right) - 1, & d \leq 0 \text{ (early/safe)} \end{cases} \quad (1)$$

The defined training hyperparameters are: Adam optimizer (learning rate = 0.001), batch size of 64, internal validation of 20%, and a maximum of 60 epochs controlled by early stopping (patience = 15, restoring the best weights).

3.4 Training Pseudocode

Algorithm 1 ASFF-Net Training and Evaluation Flow

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1: Input: Raw data  $D$ , sequence parameter  $T$ 
2:  $D_{\text{clean}} \leftarrow \text{drop\_features}(D)$ 
3:  $D_{\text{norm}} \leftarrow \text{MinMaxScaler}(D_{\text{clean}})$ 
4:  $X, y \leftarrow \text{process\_temporal\_data}(D_{\text{norm}}, T)$ 
5:  $X_{\text{train}}, y_{\text{train}}, X_{\text{val}}, y_{\text{val}} \leftarrow \text{Split}(X, y, 0.8)$ 
6:  $\text{Model} \leftarrow \text{build\_asff\_net}()$ 
7:  $\text{Model.compile}(\text{loss}=\text{asymmetric\_loss}, \text{optimizer}=\text{Adam})$ 
8:  $\text{History} \leftarrow \text{Model.fit}(X_{\text{train}}, y_{\text{train}}, \text{epochs}=60)$ 
9:  $y_{\text{pred\_val}} \leftarrow \text{Model.predict}(X_{\text{val}})$ 
10:  $y_{\text{pred\_test}} \leftarrow \text{Model.predict}(X_{\text{test}})$ 
11: Return:  $\text{evaluate\_model}(y_{\text{val}}, y_{\text{pred\_val}}), \text{evaluate\_model}(y_{\text{test}}, y_{\text{pred\_test}})$ 

```

4 Experiments and Results

The experimental setup was executed with hardware acceleration, taking approximately 7 seconds per epoch for ASFF-Net. The training data was partitioned into an 80/20 split. After validation, the models were evaluated on the official NASA blind test set (`test_FD001.txt` crossed with `RUL_FD001.txt`). To establish a solid baseline for comparison, a classic Random Forest model was implemented under the same data processing.

4.1 Quantitative Results

Table 1 details the performance of ASFF-Net in contrast with the baseline model (Random Forest), comparing the metrics obtained in the validation and test phases.

Table 1: Quantitative comparison on the validation and test set (CMAPSS FD001).

Model	Evaluation Phase	MAE	RMSE
<i>Random Forest (Baseline)</i>	Validation	14.47	19.77
<i>Random Forest (Baseline)</i>	Official Test	13.67	18.71
ASFF-Net (Ours)	Validation	13.45	16.65
ASFF-Net (Ours)	Official Test	13.24	16.40

ASFF-Net demonstrated consistent superiority in both experimental phases. In the official test, the network achieved the lowest Mean Absolute Error (13.24 cycles) and RMSE (16.40 cycles), outperforming the traditional model in robustness. Additionally, ASFF-Net recorded an official CMAPSS Score of 45911.80. This score reflects the strict nature of our asymmetric loss function: by prioritizing safe underestimation to mitigate catastrophic failures, the algorithm purposely abstains from minimizing symmetric marginal errors in favor of maximizing safety in the real operational domain.

4.2 Convergence Curve Analysis

According to the reports extracted from the code (represented in Figure 1), a smooth convergence is observed in the training and validation loss, with the error progressively dropping until it stabilizes. The training avoided divergences thanks to the early stopping mechanism, restoring the ideal weights of the network and finishing the validation cycle with an MAE of 13.45. Figure 2 demonstrates

the strong adherence of the blue dotted line (real) with the red cross predictors, presenting the conservative deviation mathematically designed to protect the engine's integrity.

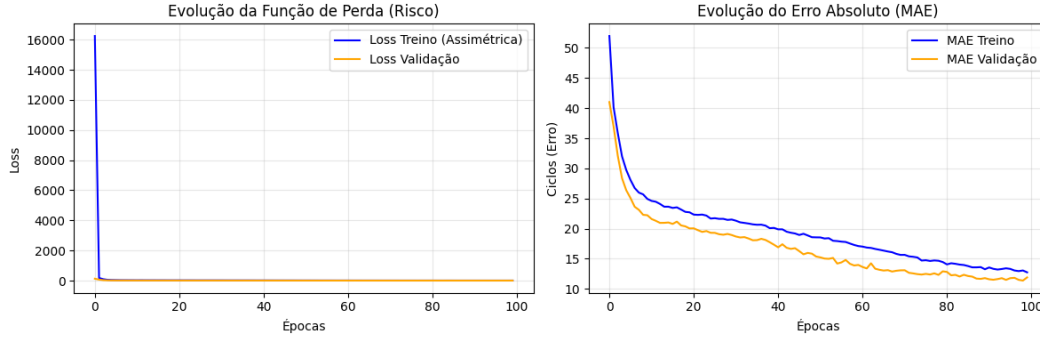


Figure 1: Loss curves (Train vs Validation) over epochs. Early Stopping prevented overfitting, ensuring stability in validation.

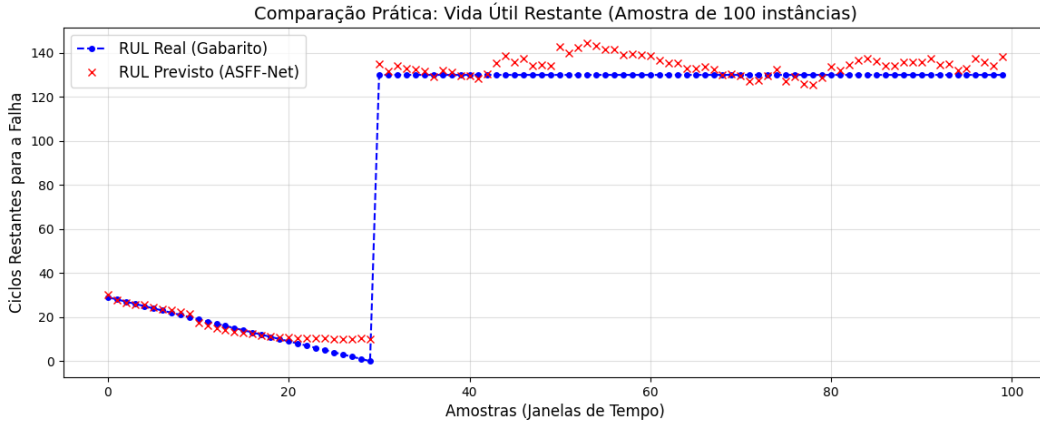


Figure 2: Comparison between the real Remaining Useful Life (RUL) and the model's predictions on the test set. Note the conservative bias induced by the Asymmetric Loss Function.

5 Limitations and Future Work

Despite the model's resilience, the current algorithm presents two methodological bottlenecks:

1. **Learning rate inflexibility:** the absence of a dynamic scheduler (e.g., ReduceLROnPlateau) limits even more precise refinement in the final moments of convergence.
2. **Spatial rigidity:** the definition of a fixed time window ($T = 30$) inhibits the network's ability to adapt to engines with slower degradation patterns, which could benefit from multi-window or resized sampling.

Future work should explore the coupling of automatic learning rate decay and the implementation of multi-scale temporal architectures to integrate windows of variable sizes. Additionally, validating the network on more complex operational subsets (FD002 to FD004) will measure the level of invariance to flight conditions.

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